

39.6 Summary and Notes

39.6.1 Summary

Idea.

1. A response family is chosen from the support and from the mean–variance relationship; with replicates, the variance-function power is estimated by regressing log group variance on log group mean (Proposition 39.1.1(a)).
2. Additivity belongs to the link, not to the data: a mean additive on one link scale is additive on another only when the two differ by an affine map (Proposition 39.1.1(b)).
3. Likelihood criteria compare models for the same random variable. A model fitted to $t(\mathbf{y})$ needs the correction $-2 \sum_i \log t'(y_i)$, itself of order n , before its AIC can stand beside one for $\mathbf{y}$ (Proposition 39.1.1(c), Example (How long does a strike last)).
4. Diagnostics should be aimed. Pearson residuals have approximate variance $1 - h_{ii}$; a simulated envelope has pointwise level $2/(B + 1)$; the link is tested by adding $\hat{\eta}^2$; a covariate's scale by partial-residual and added-variable plots; the variance function by the height and trend of $|r^P|$ (Proposition 39.2.1). In Example (Days of television news) only the last is wrong.
5. The log-likelihood is quadratic in $\boldsymbol{\beta}$ for every design exactly when the family is normal (Proposition 39.3.1): in the Bayesian thread every definition survives, every closed form does not.
6. The Bayesian generalized linear model is Definition 39.4.1. A proper prior with a bounded likelihood gives a proper posterior; a flat prior on a logistic model gives one exactly when the data are in overlap, that is exactly when the maximum likelihood estimate exists (Proposition 39.4.1).
7. Under local smoothness, a growing Hessian and a concentration condition, the Laplace approximation to the marginal likelihood has relative error $O(n^{-1})$ (Theorem 39.4.2); its logarithm is the BIC with an $O(1)$ remainder carrying the whole influence of the prior (Corollary 39.4.3).
8. Metropolis–Hastings and Gibbs leave the posterior invariant by detailed balance (Proposition 39.4.4, Definition 39.4.2); Albert–Chib augmentation makes every probit full conditional normal or truncated normal (Proposition 39.4.5). Convergence is judged by $\hat{R}$, mixing by effective sample size (Example (The 1996 survey, sampled)).
9. Posterior predictive checks compare the data with replicates from Eq. 39.4.8; using the data twice makes them conservative, so an extreme p -value is evidence and a middling one is not (Definition 39.4.3, Example (Where the survey model fails)).
10. A normal prior is ridge, a Laplace prior the lasso at its mode, and a normal prior on a logit is bimodal on the probability scale exactly when $\sigma^2 > 2$ (Proposition 39.5.1). A proper prior makes the mode exist under separation, where maximum likelihood has none (Example (The separated subgroup, three priors)).

39.6.2 Notes and sources

Model building. The three-part structure — random component, systematic component, link — is Nelder and Wedderburn's (1972); McCullagh and Nelder (1989) remains the standard

account of how the pieces are chosen, their chapter 8 for the log link being preferred to the canonical reciprocal in gamma models and their chapter 11 for mean–variance diagnostics. Fahrmeir, Kneib, Lang and Marx (2021, §5.3) develop the log-normal and gamma alternatives for nonnegative responses. The Jacobian correction of Proposition 39.1.1(c) is elementary, widely known, and as widely omitted from software output.

Diagnostics. Pregibon (1981) transferred the linear-model diagnostics of Chapter 20 to generalized linear models by the substitution $\mathbf{X} \mapsto \hat{\mathbf{W}}^{1/2}\mathbf{X}$, and Pregibon (1980) introduced the goodness-of-link test of Proposition 39.2.1(d). The half-normal plot with a simulated envelope is Atkinson’s (1981, 1985), and the exchangeability argument behind Eq. 39.2.1 is the one that justifies Monte Carlo tests generally. Cook and Weisberg (1983) read a trend in absolute residuals as evidence about the variance function.

The Bayesian thread. Lindley and Smith (1972) is the origin of the hierarchical linear model. Diaconis and Ylvisaker (1979) characterized the conjugate priors of an exponential family; the generalized linear form Eq. 39.3.2, with a pseudo-response and per-observation weights, is Bedrick, Christensen and Johnson (1996) and Chen and Ibrahim (2003). Rencher and Schaalje (2008, chapter 11) and Agresti (2015, chapter 10) are the source treatments closest to this chapter, and Gelman, Carlin, Stern, Dunson, Vehtari and Rubin (2013) the standard modern reference for its last two sections.

Laplace and the BIC. Laplace’s method is two centuries old; its use for posterior moments and marginal likelihoods is Tierney and Kadane’s (1986). Schwarz (1978) derived the BIC, and Kass and Raftery (1995) made the link between Bayes factors, the Laplace approximation and the BIC standard; their section 4 discusses the regularity conditions that Theorem 39.4.2 assumes, and the set used here — a shrinking ball, bounded scaled derivatives, a concentration bound — is convenient rather than weakest. The deviance information criterion is Spiegelhalter, Best, Carlin and van der Linde (2002); the Bernstein–von Mises theorem in the form quoted is van der Vaart (1998, chapter 10).

MCMC. The algorithm of Eq. 39.4.6 is Metropolis, Rosenbluth, Rosenbluth, Teller and Teller (1953) as generalized by Hastings (1970); Tierney (1994) gave the rigorous account for continuous state spaces and Robert and Casella (2004) the textbook treatment. The Gibbs sampler is Geman and Geman (1984), brought into mainstream statistics by Gelfand and Smith (1990). Optimal scaling is Roberts, Gelman and Gilks (1997), and the iteratively reweighted proposal follows Gamerman (1997). Data augmentation for the probit model is Albert and Chib (1993); for the logit, Holmes and Held (2006) use a scale mixture and Polson, Scott and Windle (2013) the Pólya–gamma identity now in general use. Eq. 39.4.7 is Gelman and Rubin (1992) and the truncation rule for the effective sample size is Geyer’s (1992). Posterior predictive checking is Rubin (1984), Meng (1994) and Gelman, Meng and Stern (1996), its p -values analysed by Bayarri and Berger (2000). Hamiltonian Monte Carlo, which is what modern software actually runs, is omitted because its justification needs machinery this book does not develop.

Priors as penalties. The ridge–normal correspondence is as old as ridge regression (Theorem 27.2.2); the lasso–Laplace one was noted by Tibshirani (1996) and turned into a sampler by Park and Casella (2008). Gelman, Jakulin, Pittau and Su (2008) proposed the Cauchy default prior for logistic regression with the rescaling to standard deviation 1/2, and gave the separation argument that Proposition 39.5.1(d) formalizes. That Firth’s (1993) penalty is the Jeffreys (1946) prior was observed by Firth himself; Heinze and Schemper (2002) made it the standard remedy. The horseshoe is Carvalho, Polson and Scott (2010). The propriety argument of Proposition 39.4.1(b) is the Bayesian face of Theorem 35.3.1.

39.6.3 References

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